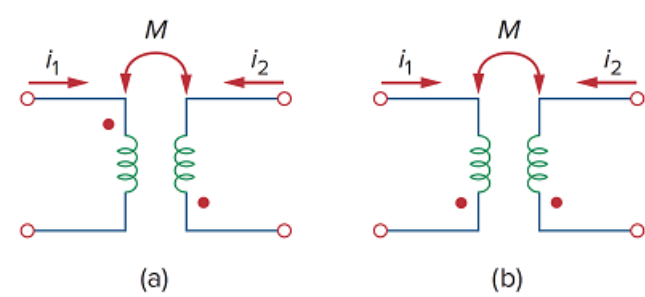


## Problem

Refer to the two magnetically coupled coils of Fig. 13.69(a). The polarity of the mutual voltage is:

- (a) Positive
- (b) Negative

**Figure 13.69**



## Step-by-step solution

## Step 1 of 2

Refer to Figure 13.69(a) in the textbook.

Recall the statement:

If a current leaves the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is negative at the dotted terminal of the second coil.

## Step 2 of 2

Observe the magnetically coupled coils in Figure 13.69(a).

Here, the current leaves the dotted terminal of one coil.

Therefore, the polarity of the mutual voltage is **Negative**.

Hence, the correct option is **b**.